Polynomial Regression

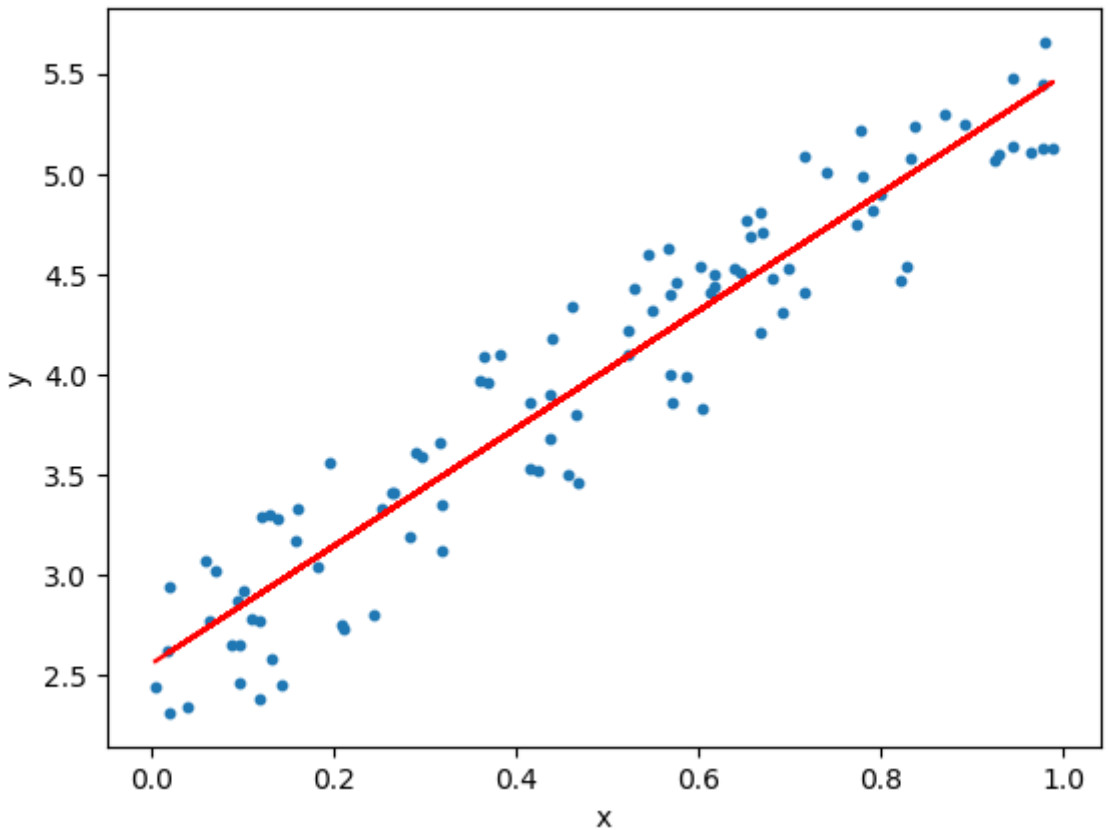
Simple Linear Regression

In Linear Regression, the relationship between the dependent and independent variable is linear

But here you have one dependent and one independent variable.

Equation

$$Y = \beta_0 + \beta_1 X + \epsilon$$



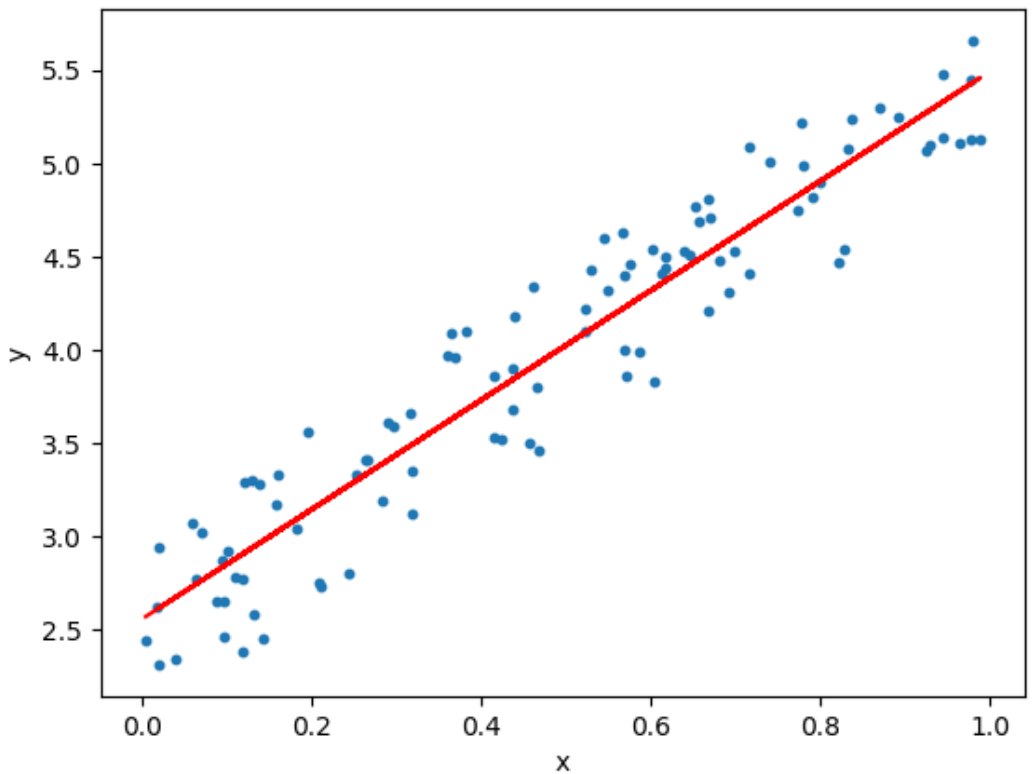
MultiLinear Regression

In Linear Regression, the relationship between the dependent and independent variable is linear

But here you have one dependent and multiple independent variable.

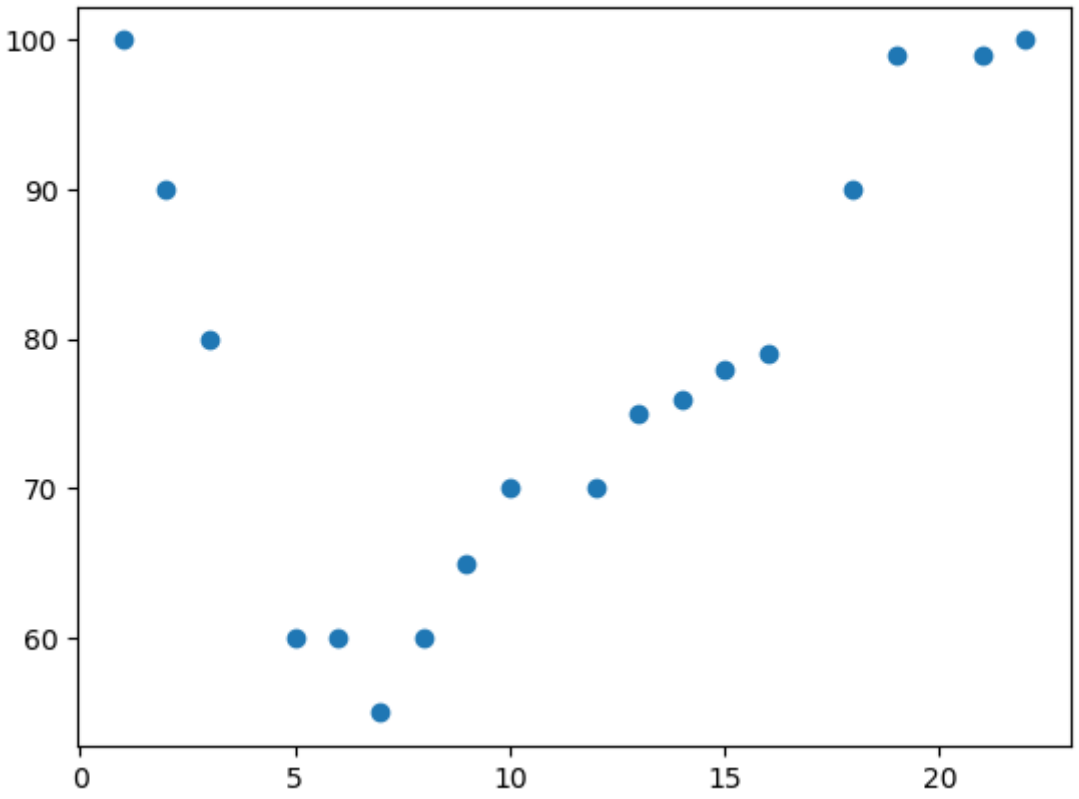
Equation

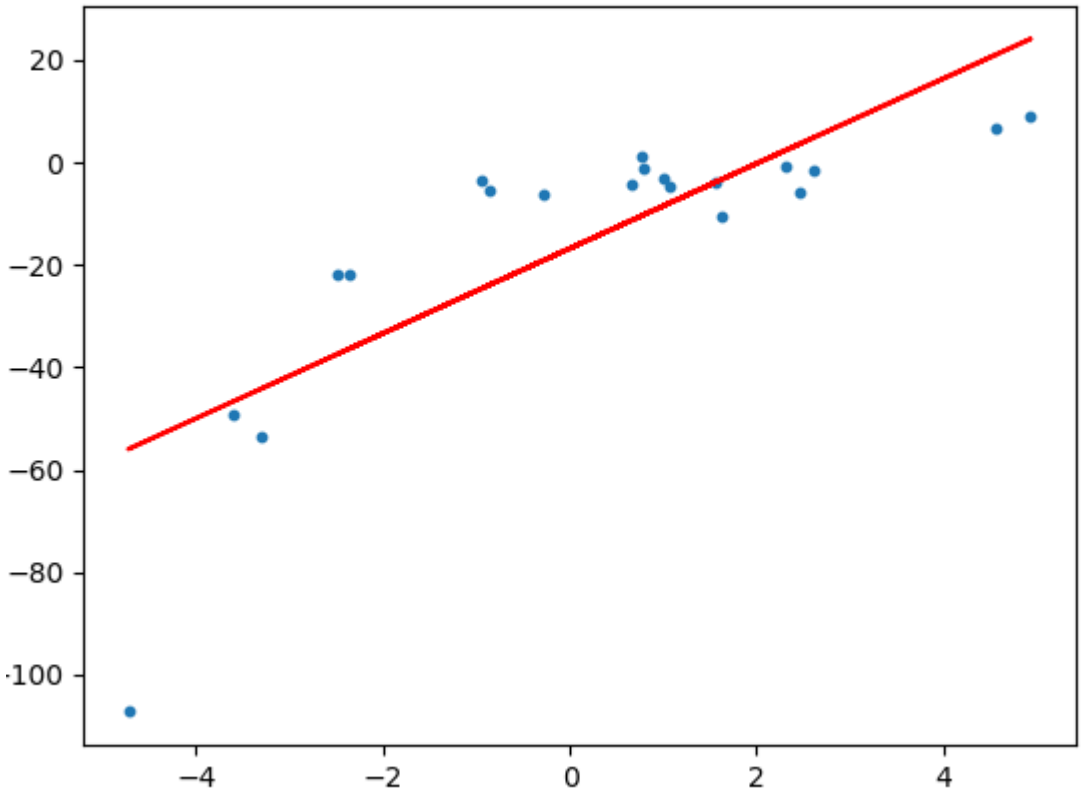
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots \ldots \ldots \ldots + \beta_n X_n + \epsilon$$



Complex Data:

Some times the data or the distribution of the data is more complex.
The data may not be Linear.
The Linear models we have seen will not be able to fit non linear data.



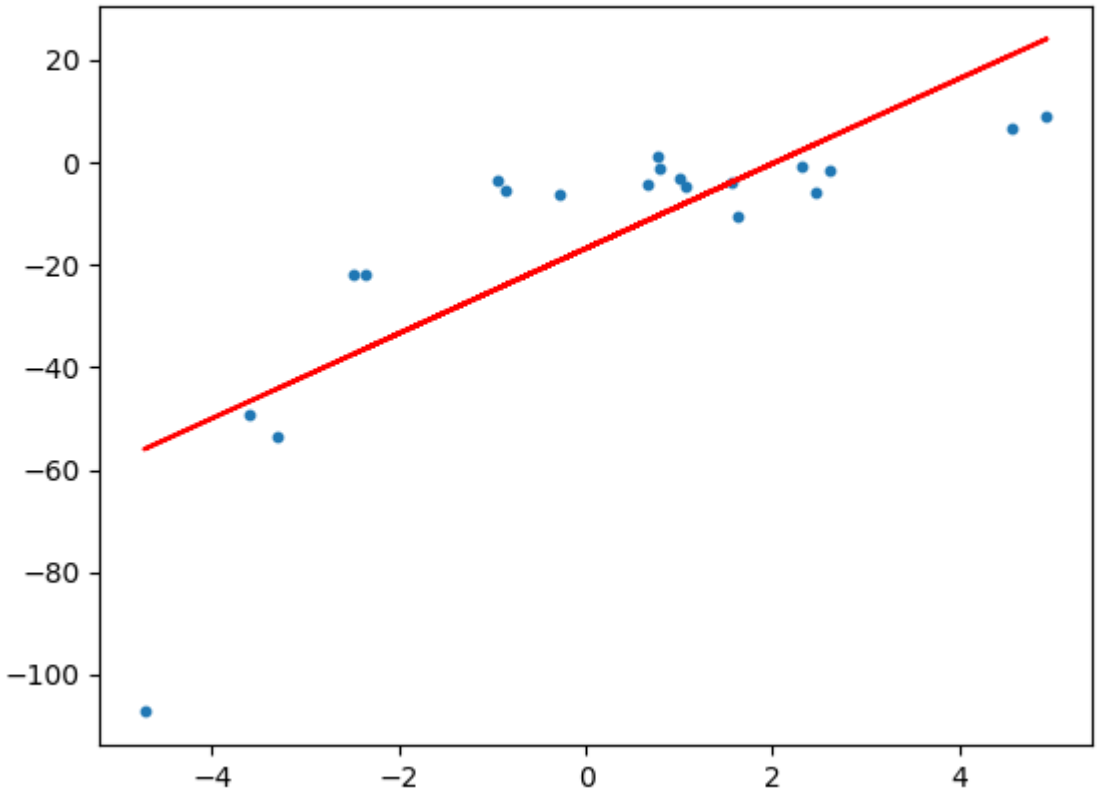


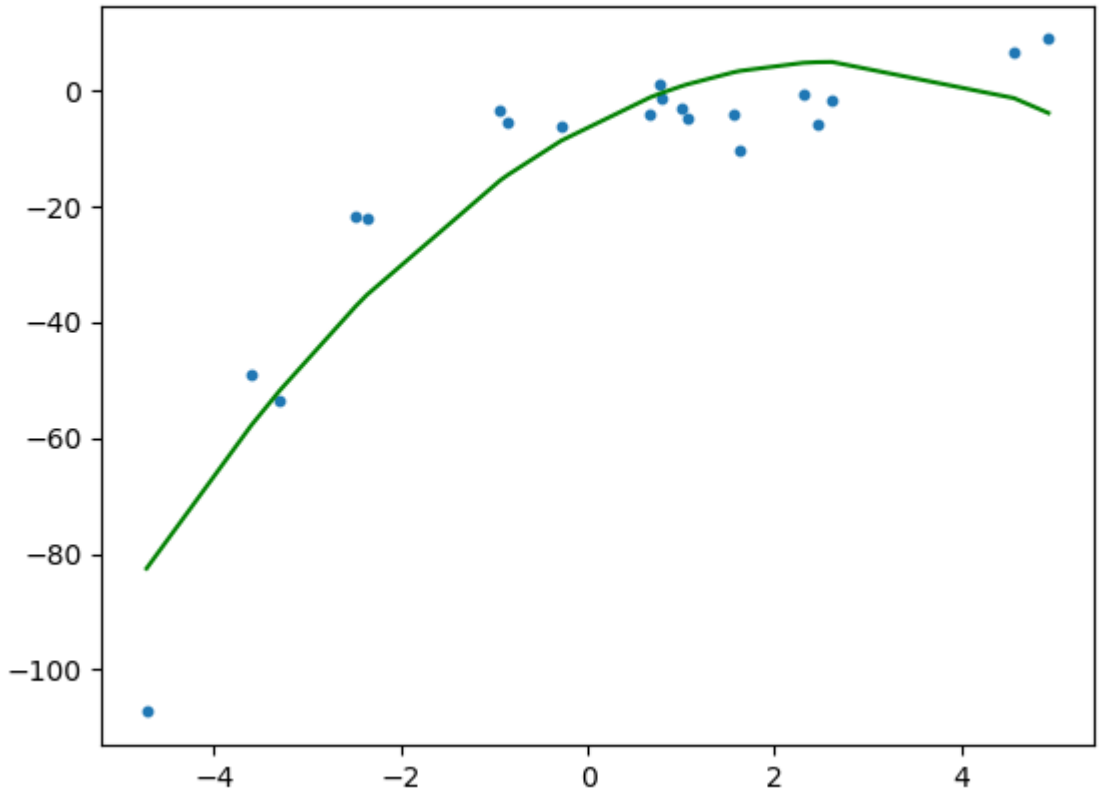
The above line under fits the data points.

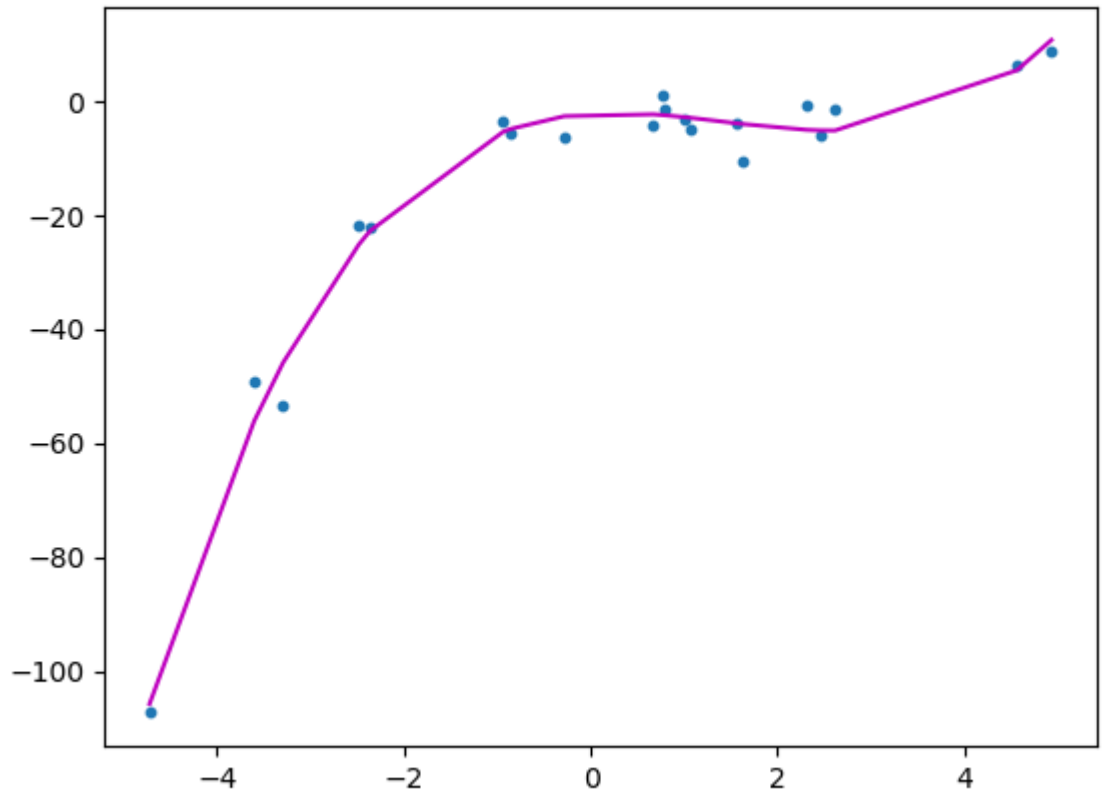
Polynomial Regression:

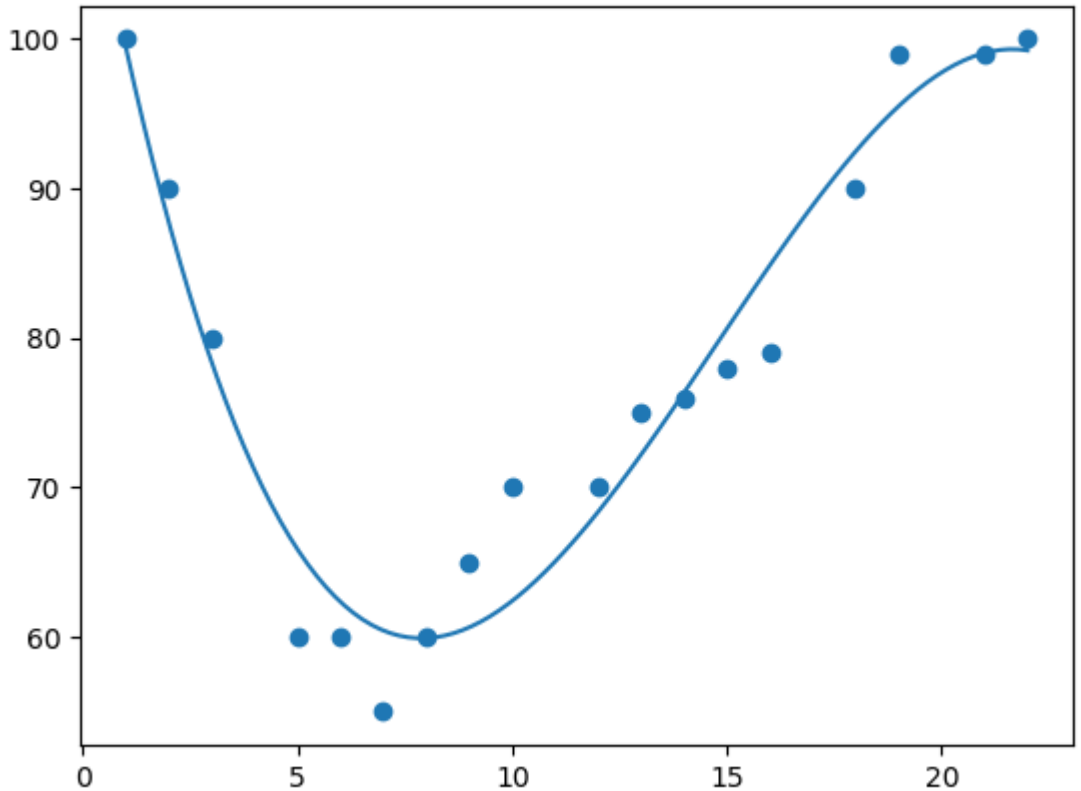
Polynomial regression, like linear regression, uses the relationship between the variables x and y to find the best way to draw a line through the data points.

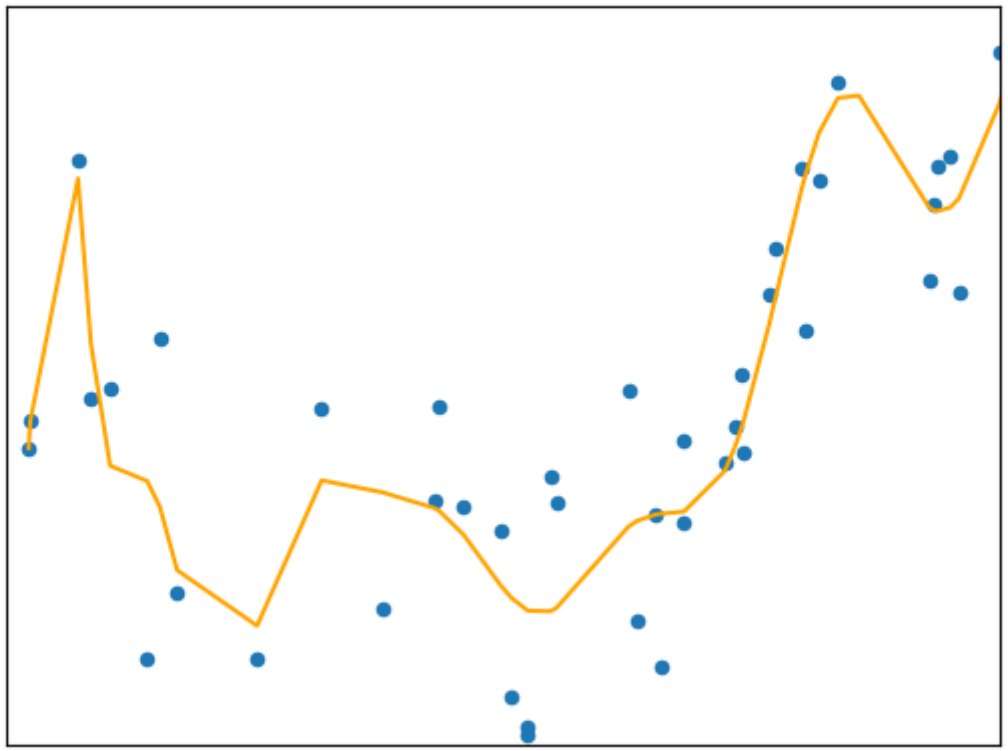
So instead of a straight line we require a curve fitting line.




0
−20
−40
−60
−80
−100
−4
−2
0
2
4


0
−20
−40
−60
−80
−100
−4
−2
0
2
4





It is clear from the above plot that the quadratic curve is able to fit the data better than the linear line.

To overcome under-fitting, we need to increase the complexity of the model.

To generate a higher order equation we can add powers of the original features as new features.
Therefore the linear model,

$$Y = \beta_0 + \beta_1 X$$

Can be transformed to

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \cdots \ldots \ldots + \beta_n X^n + \epsilon$$

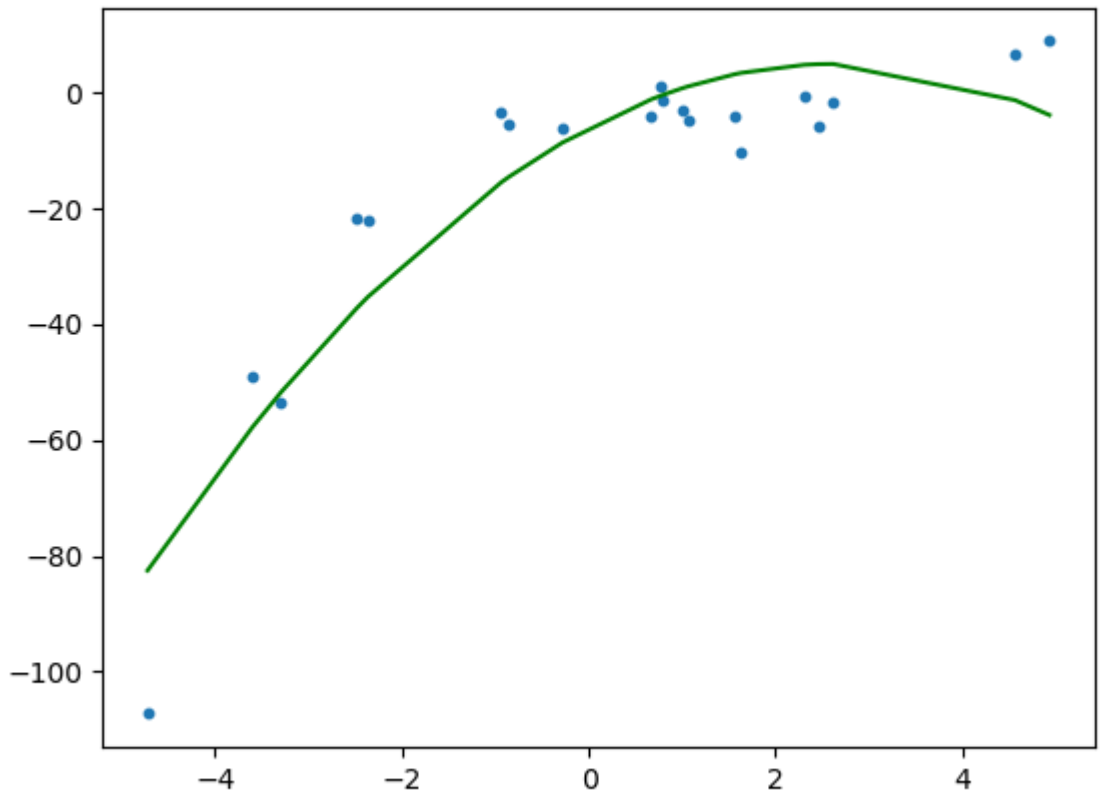
The order can be 2 or 3 or 4, so on till n.
But the most we will go for is the 2nd order.
This is still considered to be **linear model** as the coefficients/weights associated with the features are still linear.
x^2 is only a feature.
However the curve that we are fitting is **quadratic** in nature.

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

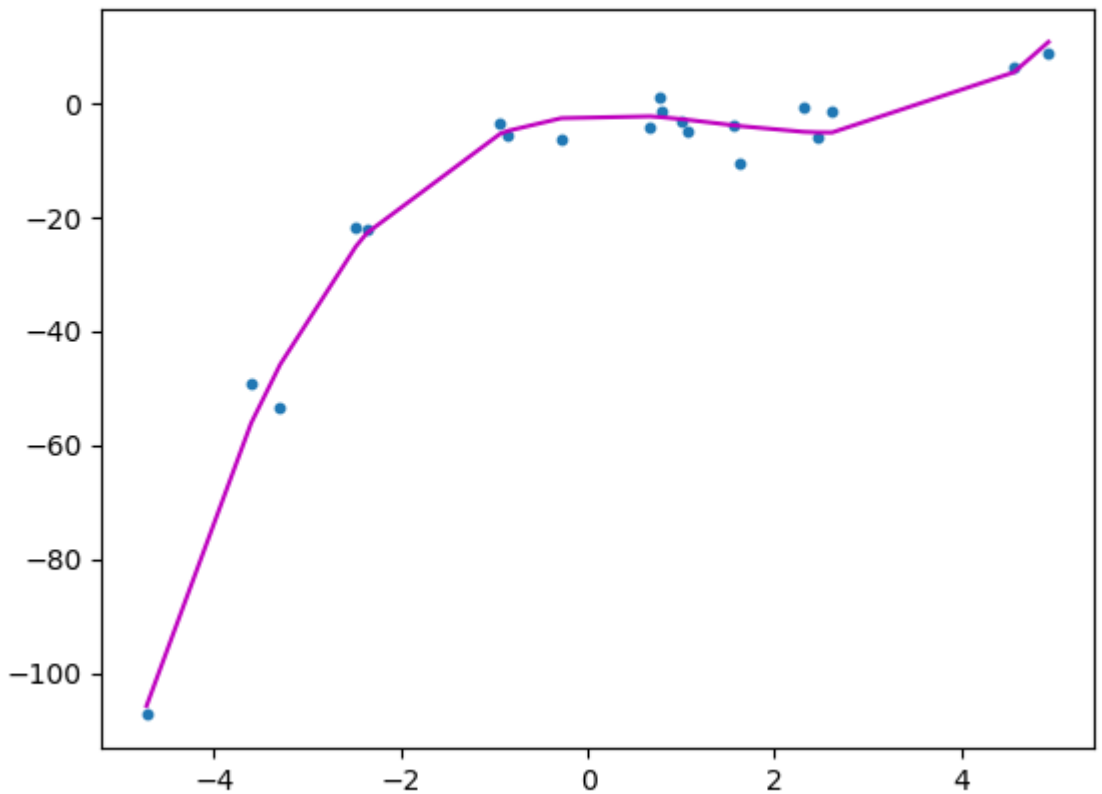
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However the curve that we are fitting is **quadratic** in nature.
It can be cubic also.
It can be 15 or 20 and so on.

2nd Order: Quadratic

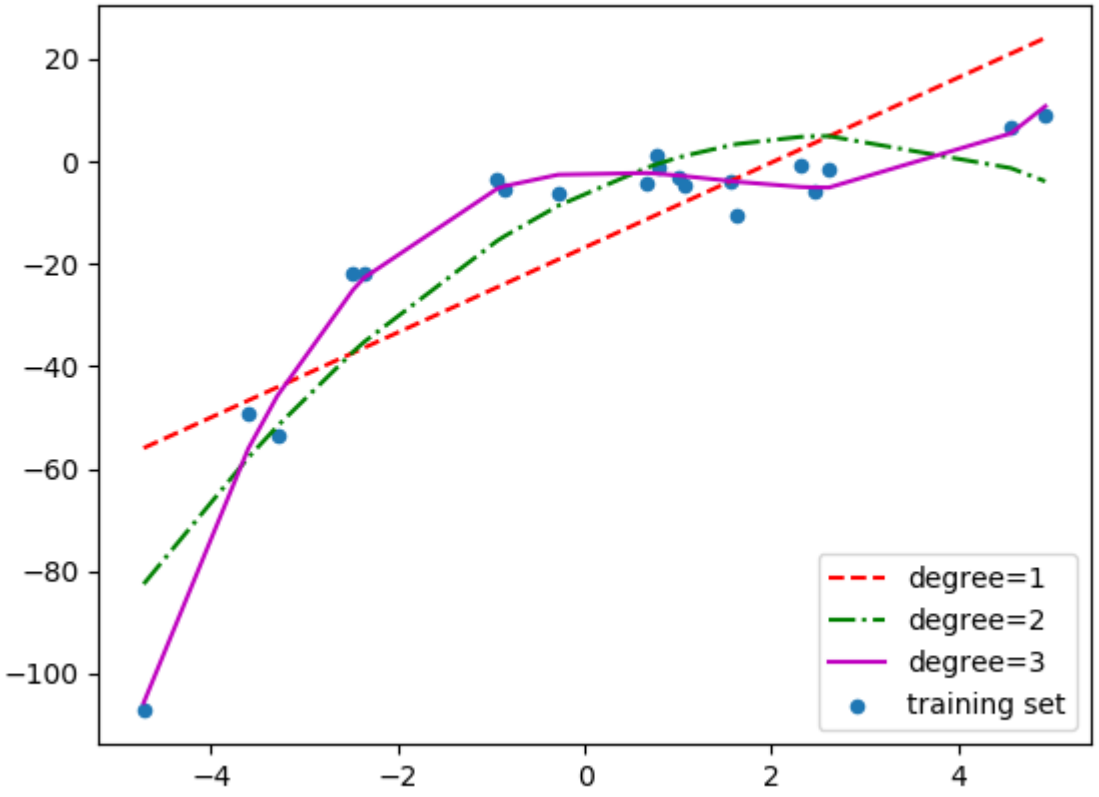
$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$



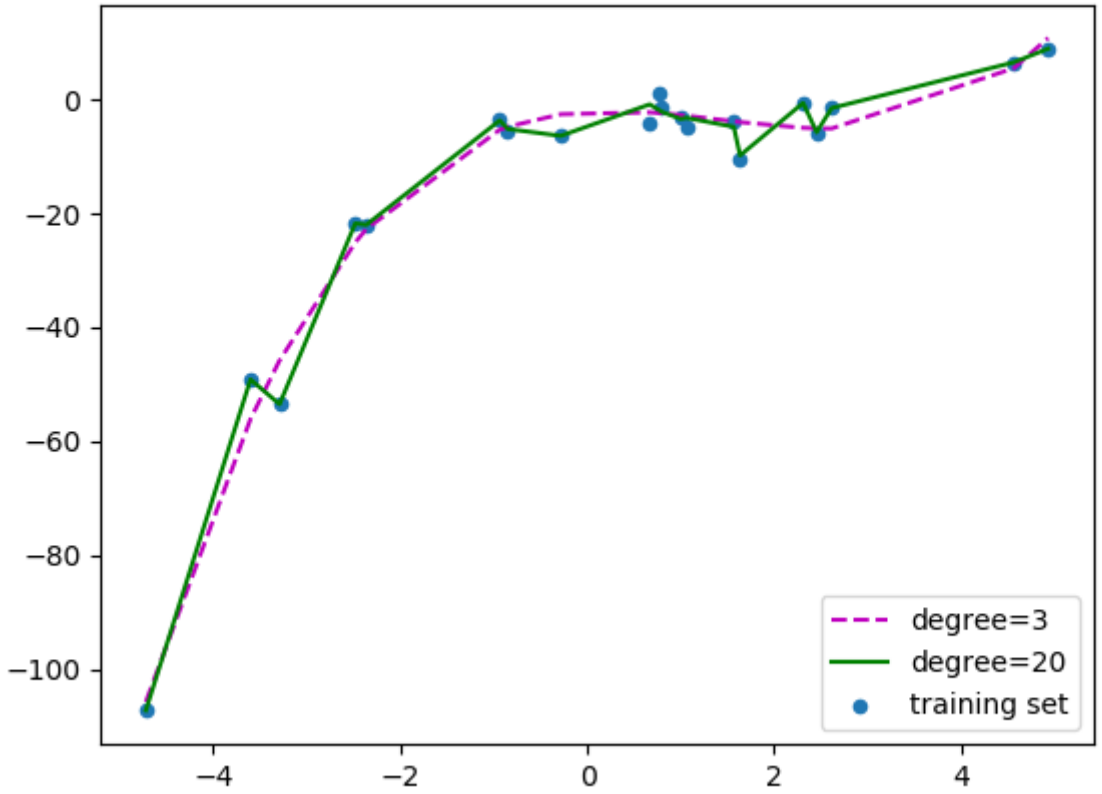
3rd Order: Cubic



Below is a comparison of fitting linear, quadratic and cubic curves on the dataset.



If we further increase the degree to 20, we can see that the curve passes through more data points. Below is a comparison of curves for degree 3 and 20.



For degree=20, the model is also capturing the noise in the data. This is an example of **over-fitting**. Even though this model passes through most of the data, it will fail to generalize on unseen data.

To prevent over-fitting, we can add more training samples so that the algorithm doesn’t learn the noise in the system and can become more generalized.

It is always preferred to have the order of the polynomial to be less than or equal to 2.

Equation:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_m x_i^m + \varepsilon_i \ (i = 1, 2, \ldots, n)$$

can be expressed in matrix form in terms of a design matrix $\mathbf{X}$, a response vector $\vec{y}$, a parameter vector $\vec{\beta}$, and a vector $\vec{\varepsilon}$ of random errors. The *i*-th row of $\mathbf{X}$ and $\vec{y}$ will contain the *x* and *y* value for the *i*-th data sample. Then the model can be written as a system of linear equations:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ 1 & x_2 & x_2^2 & \cdots & x_2^m \\ 1 & x_3 & x_3^2 & \cdots & x_3^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^m \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{bmatrix},$$

which when using pure matrix notation is written as

$$\vec{y} = \mathbf{X}\vec{\beta} + \vec{\varepsilon}.$$

Equation:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

How to find the X coeffecients, that is

$$\beta_0, \beta_1, \beta_2 \; and \; so \; on$$

The equation is the same as Linear Regression

$$\hat{\beta} = (X^T X)^1 X^T Y$$

Once $\beta_0, \beta_1, \beta_2 \; and \; so \; on$ is known, we can fit them in the below equation

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

If more number features are there, then the equation will change to:

For example if there are 2 features X1 and X2, then the equation will change to:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_1 X_2 + \beta_2 X_2^2 + \epsilon$$

The major issue with Multivariate Polynomial Regression is the problem of **Multicolinearity**. When there are multiple re- gression variables, there are high chances that the variables are interdependent on each other.

The vector of estimated polynomial regression coefficients (using ordinary least squares estimation) is

$$\hat{\vec{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\,\mathbf{X}^{\mathsf{T}}\vec{y},$$